

MNIST

Pattern Recognition - Baseline Experiment

This demonstrates supervised learning using topological associative memory. Raw images have been preprocessed with Gaussian blur.

Image data is encoded as sparse distributed representations (SDRs), using a random sparse projection followed by k-winners-take-all.

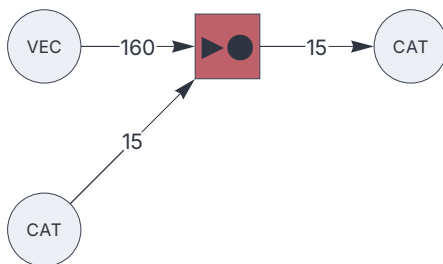
We encode labels as non-overlapping random sparse holographic representations (SHRs).

Configuration

```
<< "Circuits.m"
```

```
dataflow =
{
  input[category[10]] [11] [],
  input[vectorencoder[]] [12] [],
  associator[] [101] [11 "%", 12 "%"],
  output[category[10]] [] [101 "%"],
  12 → {16000, 160}, (* image projection *)
  _ → {150, 15}      (* labels / categories *)
};
```

```
c = Circuit[ dataflow]; c[Graph, ImageSize→ 240]
```



Dataset

```
labeleddata = Import[NotebookDirectory[] <> "/data/MNIST-blurred.wxf"];  
Length /@ %
```

```
{70 000, 70 000}
```

Train

```
{data, categories} = Take[#, 60000] & /@ labeleddata;
```

```
c @@@ Transpose[ {categories, data}]; (* Train *)
```

```
results = c[Null, #]& /@ data; (* Test with training data *)
```

```
100. Mean@Boole@MapThread[SameQ, {results, categories}] (* Score *)
```

```
99.9983
```

Test

```
{data, categories} = Take[#, -10000] & /@ labeleddata;
```

```
results = c[Null, #]& /@ data; (* Test with test data *)
```

```
100. Mean@Boole@MapThread[SameQ, {results, categories}] (* Score *)
```

```
96.04
```